

Technology for Success: Computer Concepts By Module Overview of Critical Thinking Activities

WHAT IS IT FOR? Summarizes each narrative module's objectives and the corresponding non-narrative activities to use for critical thinking exercises: Critical Thinking Challenges (CTCs) and Technology for Success Concept Videos.

RECOMMENDED USE: Instructors can use this brief overview to help plan their assignments (e.g. selecting the modules that they want to cover or the sequence).

Module	Lesson Objective	Critical Thinking Challenge, available in MindTap & SAM	Concept Video, Skills Training, and Exam, available in MindTap & SAM
1: Impact of Digital Technology	Develop personal uses for technology to help with productivity, learning, and future career growth. In this simulation the student must decide: - How/when to integrate technology into their daily life - How to use technology to enhance their productivity and efficiency	Use Technology in Daily Life - Technology helps save us time and accomplish our goals. - Think about which technology options you might choose on your way to class. - You have several tasks on your to-do list before class. - You need to deposit your check. - Your rent is due - You need to buy books for school	- Assist users with limited mobility. - Conduct a job search online - Explore the impact of virtual reality. - Find information on open sites. - Perform daily living tasks online. - Use artificial intelligence software. - Use browser extensions. - Use business-to-business e-commerce. - Use business-to-consumer e-commerce. - Use consumer-to-consumer e-commerce. - Use embedded computers in home appliances. - Use embedded computers in motor vehicles. - Use embedded computers in stores. - Use enterprise computing. - Use natural language processing. - Use phones in business. - Use productivity web apps. - Use robotics.

		before class starts this afternoon.	• Use technology in manufacturing • Use technology in transportation services. • Use technology to assist users with disabilities. • Use the Internet of Things in daily life.
2: The Web	Conduct online research. In this simulation the student must decide: • How to most effectively use search engines • How to gather information from a variety of sources • How to determine whether a website provides accurate, reliable information	Find the Perfect Video Conference Solution • You are a new employee at a virtual reality start-up company. • You will need to research the right video conferencing solution for your company.	• Access apps. • Avoid storing cookies. • Compare the pros and cons of web apps. • Define the internet • Evaluate search results • Explain how search engines work. • Following search guidelines. • Identify popular web apps. • Make E-commerce payments • Protect against online fraud. • Use a browser. • Use a search site to find information. • Use advanced search forms. • Use digital deals and discounts. • Use e-commerce in personal transactions. • Use keyboards. • Use search engine tools. • Use specialty search engines.
3: Computer Hardware	Categorize various types of computer hardware. In this simulation the student must decide: • Which type of computer and storage system is best for them • Which hardware features meet their needs	Selecting a Computer • You've been awarded a scholarship to be used for purchasing technology equipment for college courses. • You're hoping to be able to buy a computer and printer.	• Avoid extreme temperatures. • Buy a computing device • Buy add-on devices. • Define operating systems. • Define processor cache. • Define random access memory • Define software. • Define system software. • Describe benchmarking.

		You have to report back within 30 days on how you spent the money.	- Describe laptops. - Edit video files. - Evaluate hardware requirements. - Explain how computers represent data. - Identify communications hardware. - Identify types of printers. - Install hardware. - Install software. - Maintain removable media. - Provide voice input. - Record sound. - Restore an operating system. - Store data in the cloud - Store data on a network. - Store data on an internal hard drive. - Store files. - Troubleshoot peripheral device problems. - Troubleshoot storage device problems. - Use cameras and scanners. - Use microphones. - Use pen input devices. - Use pointing devices. - Use speakers and headphones. - Use touch screens. - Use voice output.
4: Operating Systems and File Management	Manage files and folders. In this simulation the student must decide: How to best save and organize files	Working with OneDrive Files You're working as an office assistant on a college campus, and you've been asked to help finalize a poster to	- Create and rename folders. - Describe processing techniques. - Explain the differences between the major operating systems. - Identify an operating system. - Identify mobile operating systems - Identify PC operating systems.

	• How to share local and online files • How to most effectively name files	advertise an upcoming annual fundraising event. • You've been instructed to coordinate with the sponsoring company to make sure the poster is final before it is printed.	• Identify the key features of mobile apps. • Manage input and output • Move, copy, and delete folders. • Navigate folders. • Open a saved file. • Open windows. • Reduce electronic trash. • Resize windows. • Run disk utilities. • Run file utilities. • Save a file to a location on the hard drive. • Select a PC operating system. • Select, copy, and move files. • Set notifications. • Set up user accounts • Shut down devices. • Solve pointing device problems • Use menus and dialog boxes. • Use virtual machines.
5: Software and Apps	Use common features of productivity apps In the simulation the student must decide: • How/when to use Access to gather data • How to present data visually • How to prepare a presentation	Working with Applications • You're working as a staff accountant for a company that owns and manages apartment complexes. • One of your responsibilities is to reconcile payroll expenses. • You need to gather data and present the results and your strategy to the management team.	• Access online education. • Access online entertainment and media. • Access online news and information. • Access videos on the web. • Access web apps. • Analyze spreadsheet data • Attend video conferences. • Compare native apps and web apps. • Describe drawing apps. • Describe paint apps. • Describe photo-editing apps. • Describe types of apps. • Design effective presentations. • Identify common features of apps. • Identify smartphones

			• Identify the key features of database software. • Identify the key features of graphics software. • Identify trends in app development • Manage databases. • Play sounds and music. • Play video. • Summarize current trends in app development. • Update apps. • Use app stores.
6: Security and Safety	Use protective measures to safeguard security and data. In this simulation the student must decide: • How to create a secure password • How to safely connect to the Internet • How to protect themselves from security attacks	Privacy and Security • You are a new sales executive for MusicX, a music subscription service for businesses. • You will meet with your first potential customers at a local coffee shop where you will navigate your new laptop and make decisions related to privacy and security in a public space.	• Address spoofing. • Describe common cybersecurity attacks. • Donate computer equipment. • Encrypt personal data. • Identify behavioral health risks. • Identify toxic electronic parts. • Identify unsafe websites. • Protect device hardware • Protect mobile devices. • Provide biometric input. • Recycle electronics. • Secure computing devices. • Understand the risks to physical, behavioral, and social health. • Use browser plug-ins. • Use common sense to secure data.
7: Digital Media	Record and edit digital media. In this simulation the student must decide: • How to acquire, license, and distribute music	Working with Media Files • You volunteer at a local recreation center and are helping to organize a	• Capture video. • Create graphics. • Define digital video. • Define file formats. • Define graphics software.

	• How to create or acquire photos • How to edit and post videos	fitness program for students. • Since you have an associate's degree and work in a technology field, you volunteer to coordinate the technical side of creating the multimedia resources. • You've been given access to a computer with several media editing apps and tools already installed to get you started.	• Define resolution and compression. • Download graphics. • Edit sound files. • Explain how to create graphics and animation. • Participate in mobile gaming. • Set up a gaming system. • Stream audio. • Stream television shows. • Stream videos • Use animation on the web. • Use animation. • Use computer-aided design. • Use game controllers. • Use icons and buttons. • Use synthesized music. • Use synthesized speech.
8: Program and App Use and Development	Describe the methods associated with developing software. In this simulation the student must decide: • Whether to develop an installed mobile app or web-based app • How to manage user accounts • What features to include • Whether to use a programmer or designer to develop the app or website	Creating a Successful App • You and two friends run a sports gear store that specializes in bicycles and need a way to stay ahead of the competition. • One idea you've had is to develop an app. • You've decided to invest some money into hiring a software developer to create the app. You must consider how you'll deliver your app to your	• Access restricted sites. • Combat software piracy. • Define the key terms associated with object-oriented programming. • Identify development methodologies. • Install service packs and updates. • Recover from software crashes. • Troubleshoot software problems. • Use a firewall on a network. • Use embedded software. • Use freeware, shareware, and public domain software. • Use license agreements. • Use preinstalled software. • Use programming languages.

		customers.	Use programming tools.
9: Web Development	Explain strategies for creating and publishing websites. In this simulation the student must decide: - Whether to hire a web designer or use DIY tools to renovate the website - What features to include - Whether to use a template or to build the site of scratch - Which html tag to use to fix a problem	Renovating Your Company Website - You manage the daily operations of Beeman & Riddle Investments. - The owners ask you to review the website and make suggestions for improvement and updates in order to increase website traffic and bring in new customers. - You have explored the existing website and need to make decisions regarding the website renovation.	- Define a graphical user interface - Define word processing software. - Explain how to use HTML. - Identify recent top-level domains. - Manage disk space.
10: Networking	Explore key features of connected networks. In this simulation the student must decide: - The best uses for various network packages - How to securely set up a network - How to securely connect to a network	Determining Network Requirements - It is your first day at Chilli Networks, and you are making your first customer calls. - Your job is to help your customers find the right	- Access cellular networks. - Compare intranets, extranets, and VPNs. - Define cellular standards. - Define hubs, switches, and routers. - Define malware. - Describe net neutrality - Describe the pros and cons of network types. - Explain how a network operates. - Explain how to encrypt a network.

		type of network and help them use it.	- Identify local area networks. - Identify network architecture. - Identify network hardware and software. - Identify personal area networks. - Identify types of malware. - Install a local firewall.
11: Digital Communication	Demonstrate familiarity with using and creating multiple types of digital communication. In this simulation the student must decide: - How to strategically use social media to reach customers - How to create an email campaign	The DNA of a Marketing Webinar - You work on the marketing team at an innovative tech company. - You must leverage digital marketing to advertise the company's latest products. - You have already identified several ways to use digital communications to meet your goals. - You will explore how best to use various social media platforms and discover other ways you can use digital media in marketing.	- Attend webinars. - Create blogs. - Create email addresses. - Create email messages. - Define digital communication. - Define electronic mail. - Define electronic messaging. - Define social networks - Describe common types of digital communication. - Evaluate social networking. - Exchange instant messages. - Exchange messages on social networks. - Exchange text messages. - Explain the importance of netiquette. - Follow guidelines for blogs. - Follow guidelines for personal social networking. - Follow guidelines for professional messaging. - Follow guidelines for professional social networking. - Gather support with social media. - Identify types of social media. - Identify types of social networks. - Identify types of wikis.

			• Identify typical features of email software. • Use email systems. • Use microblogs. • Use podcasts.
Module 12: Cloud Computing: AI, eCommerce, and the Cloud	Explain the basics of cloud computing, describe ways companies do business on the Internet, and analyze ways to communicate more proficiently with AI systems. In this simulation, the student must decide: • How to choose business-appropriate updates to a website • How to build trust through an ecommerce website • How to interact effectively with AI	Upgrade a B2B Website • You work for a company that would like to convert their ordering system to a web-based solution. • Your job is to create a proposal to update the website.	• Identify characteristics of cloud computing technology. • Describe common cloud computing models. • Describe major cloud providers and services. • Compare various e-commerce platform types. • Describe high-growth job roles in the e-commerce industry. • Identify ways to build customer trust through a good e-commerce website. • Identify ways people use AI in everyday life. • Compare the three most common AI learning models. • Describe how AI supports smart devices and the Internet of Things. • Explain cloud technology's effect on common tasks. • Describe omnichannel marketing. • Identify AI in the workplace. • Phrase effective AI commands.
Module 13: Databases	Use a database management system. In this simulation, the student must decide:	Expand a Small Business's Database • You work for a small real estate firm and have been	• Compare spreadsheets and databases. • Explore types of databases. • Use Structured Query Language (SQL). • Secure a database. • Compare frontend and backend components.

	- How to organize a database table - How to configure fields in a table - How to choose a query parameter	tasked with improving the database. You will need to add a table for client information that meets the needs of the company.	- Organize data in a database. - Interact with data. - Backup and recover a database. - Explain the significance of big data. - Explore the impact of business intelligence.
Module 14: Digital Ethics and Lifestyle	Describe legal and ethical responsibilities of a digital citizen and explain how to be a responsible digital citizen. In this simulation, the student must decide: - How to respect copyright and privacy restrictions - How to present a positive digital footprint - How to properly dispose of old technology	Manage Job Responsibilities Professionally - You work for an insurance agency, and your boss wants you to undertake new job responsibilities. - You've been offered the opportunity to attend a conference in Mexico and will first try out your expanded responsibilities to see if you're a good fit for the new position. - You will explore ethical decisions and how your actions might affect other people.	- Describe how digital technology has revolutionized society. - Describe fair online content creation practices. - Develop accessible online content. - Cultivate a polite online presence. - Explain responsible tech disposal. - Define digital citizenship. - Describe legal, ethical, and moral issues. - Define digital equity and digital access. - Avoid risks of digital technology. - Describe reliable online sources. - Develop a digital wellness plan.